



arc42 Architecture Documentation - PixelShare

System: PixelShare

Document status: Completed architecture documentation for the MVP

Architecture style: Monolithic backend with a layered internal structure

Main scenario: Mobile image-sharing app for photography enthusiasts

This document describes the software architecture of PixelShare using the arc42 structure. The content is based on the PixelShare scenario and the SWARC work products for introduction, solution strategy, constraints, cross-cutting concepts, quality, building blocks, runtime and deployment.

Document Control

Item	Value
Team	PixelShare
Version	1.0
Status	Completed draft for review
Main architecture drivers	Fast MVP delivery, performance for images, security/privacy, maintainability
Main evaluation focus	Complete arc42 chapters, measurable quality scenarios, traceability, readable diagrams, risks and glossary

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1. Introduction and Goals

1.1 Requirements Overview

PixelShare is a mobile image-sharing app for photography enthusiasts and visual storytellers. Users can upload high-quality images, edit them, organize them, discover images from other users and share content within a photography-oriented community.

PixelShare is intended as a multiplatform application for iOS and Android. The MVP focuses on the core user journey: registration, login, image upload, image publication, feed/profile display, basic community interaction, Pixlr-based editing and subscription support.

Main Purpose	Main Features
Specialized platform for photography	Registration and login
Sharing visually appealing photos	Uploading and sharing images
Editing tools	Advanced image editing and Pixlr integration
Collections and user profiles	Community interaction: likes, comments, follows
Photography-oriented community	Content moderation and reporting
Multiplatform application	Subscription model with free and paid plans

Core functional requirements

1. Users can register, log in and manage a profile.
2. Users can upload images with title, description, tags and visibility settings.
3. Users can view images in feeds, profiles, image details and collections.
4. Users can like, comment on and share images.
5. Users can open an image-editing flow through Pixlr.
6. Users can use free or paid subscription options.
7. Admins and moderators can review reports and manage inappropriate content.
8. Support staff can analyze user-facing errors such as login, upload, subscription and editing problems.

1.2 Quality Goals

Priori ty	Quality Goal	Motivation and Concrete Scenario
QG-1	Performa nce and Reliabilit y	The app should function quickly, stably and with minimal errors. A user opens the Home Feed under normal conditions and the first visible images and posts are loaded within 2 seconds. Upload feedback is shown within 3 seconds.
QG-2	Privacy and Security	Users create personal accounts, upload their own images and may use paid subscriptions. A non-logged-in user who tries to upload an image or write a comment is immediately blocked and asked to log in.
QG-3	Usability and Learnabi lity	The app should be easy to use and quickly understandable. A new user registers, uploads an image and publishes it with title and description within 3 minutes without external help.

1.3 Stakeholders

Role/Name	Expectations
End User - Lisa Mueller	PixelShare should enable users to easily upload, view, share, discover, like and comment on images. In addition, a photography-oriented community should be created.
Development Team - CodeCraft Solutions	PixelShare requires clear technical requirements, a maintainable architecture and stable performance so that the app can be developed, maintained and improved.
UI/UX Designer - Clara Weber	PixelShare should be designed in a user-friendly and visually appealing way, so users can navigate the app easily and the images remain the main focus.
Support Team - HelpDesk PixelShare	PixelShare requires understandable error reports and user information so that support can help with login, upload, subscription and image-editing problems.
Content Moderator - Sophie Keller	PixelShare requires tools to review inappropriate content, handle reports and manage user behavior within the community.

2. Architecture Constraints

C_1 Budget Limitation for the MVP

- **Type:** Budget / organizational
- **Constraint:** The MVP must not cost more than 72,000 EUR. This includes development, testing, basic cloud/tooling costs and preparation for the app stores.
- **Motivation:** The budget is fixed because a start-up has limited financial resources.

C_2 Limited Effort and Timeline

- **Type:** Time / effort
- **Constraint:** The MVP must be delivered within 6 team sprints of 2 weeks each. This equals approximately 240 person-days.
- **Motivation:** The scope must focus first on core features to ensure a realistic implementation.

C_3 Platform Requirements: iOS and Android

- **Type:** Technical / product
- **Constraint:** PixelShare must be available on both iOS and Android.
- **Motivation:** The architecture must support both platforms with as little additional effort as possible.

C_4 Monolithic Architecture with a Clear Layer Structure

- **Type:** Architectural
- **Constraint:** The backend is implemented as a monolith with a layered architecture: Presentation/ API, Business Logic and Data Source.
- **Motivation:** Dependencies only flow downwards so that the monolith remains understandable and maintainable.

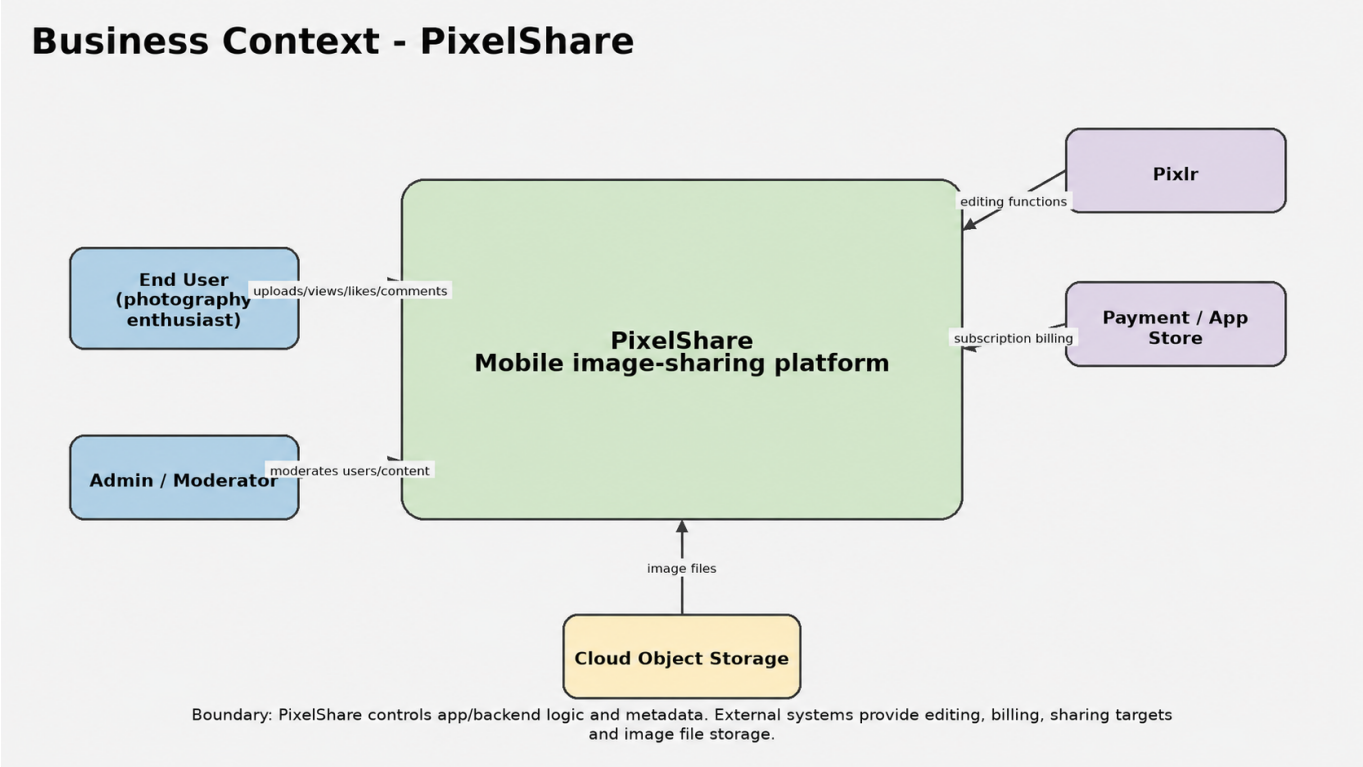
C_5 Privacy and Secure Communication

- **Type:** Legal / security
- **Constraint:** Pixlr, image storage and subscription/payment interfaces must be integrated securely. Personal data and images must comply with GDPR and privacy requirements.
- **Motivation:** PixelShare processes personal accounts, images and possible subscription data.

3. System Scope and Context

PixelShare is documented as a system that interacts with mobile users, administrators, external editing, billing providers and cloud storage targets. The core system boundary contains the mobile app, the admin interface and the backend monolith. External systems are integrated through clearly defined technical interfaces.

3.1 Business Context



Business Context Diagram

Commun-ication Partner	Input to PixelShare	Output from PixelShare
End User	Registration data, login credentials, images, titles, descriptions, tags, likes, comments, reports and sharing actions	Feeds, profiles, image details, upload results, comments, likes, error messages and share links
Admin / Content Moderator	Moderation decisions, content review actions and user-management actions	Report lists, flagged content, user information and moderation result feedback
Pixlr	Edited image callbacks or editing result information	Image or image URL for editing, edit-session request, editor configuration
Payment / App Store	Payment confirmation, subscription status and billing events	Subscription request and product/payment information
Cloud Object Storage	Stored image objects and image URLs	Image files, thumbnails and versioned image files

3.2 Technical Context

Channel / Interface	Source	Target	Protocol / Format	Purpose
Mobile API	Mobile App	API Gateway / Backend API	HTTPS/REST, JSON, multipart upload	Registration, login, feed loading, image upload, comments and likes
Admin API	Web/Admin UI	API Gateway / Backend API	HTTPS/REST, JSON	Moderation and administration
Persistence	Backend Monolith	PostgreSQL	SQL over TLS / ORM	User data, image metadata, comments, likes, subscriptions and audit data
Media Storage	Backend Monolith	Object Storage	HTTPS / S3-compatible API	Store original images, edited images and thumbnails
Editing Integration	Backend Monolith / Pixlr Adapter	Pixlr	HTTPS/API	Create edit sessions and receive edited images
Billing Integration	Backend Monolith / Payment Adapter	Payment / App Store	HTTPS/API	Subscription and billing processing
Deployment	GitHub Actions	Cloud infrastructure	CI/CD pipeline, HTTPS, secrets	Build, test and deploy the backend application

4. Solution Strategy

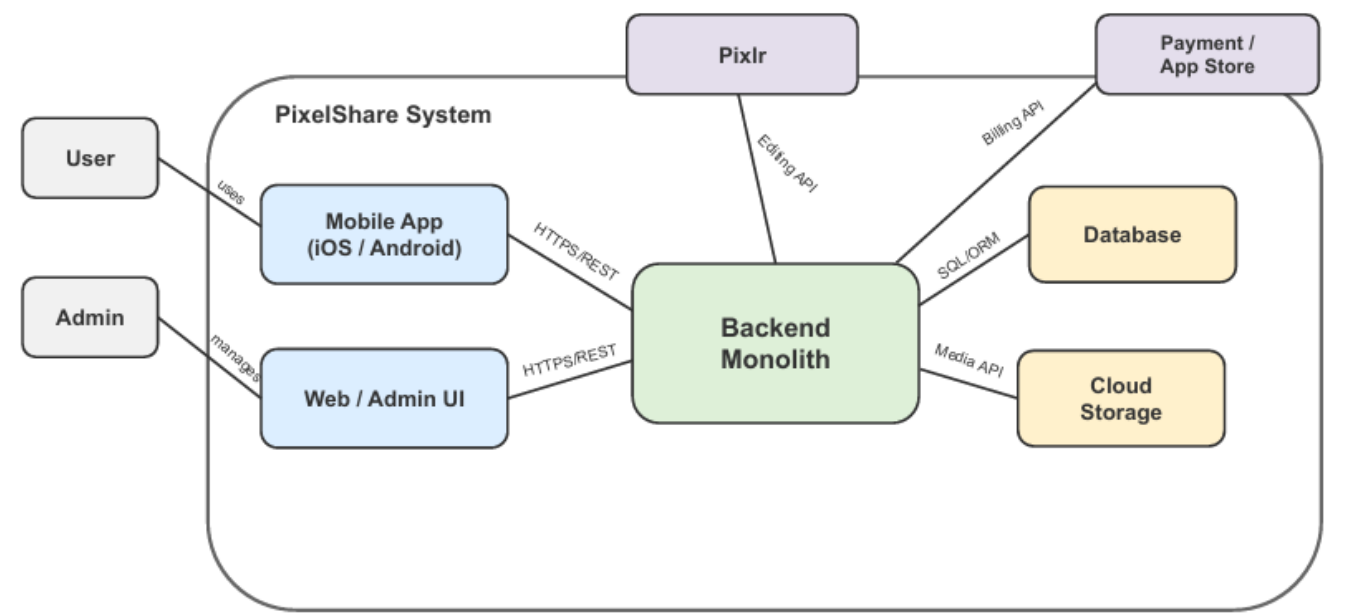
The architecture strategy focuses on delivering a reliable MVP with limited budget and team capacity. Therefore, PixelShare uses a simple and maintainable monolithic backend, clear internal layers, cloud/file storage for media files and adapter components for external providers.

Goal / Requirement	Architectural Approach
Fast MVP for a start-up	PixelShare is implemented as one central application with a single deployable backend unit. This simplifies development, testing and operation. The codebase follows clean code principles from the beginning.
Good image performance	Images are stored in object storage, while the database stores metadata such as titles, tags, likes and comments. Feeds and profiles can additionally be cached.
Secure user accounts and subscriptions	Central authentication, admin roles, encrypted passwords and secure payment processing through a payment provider are used.
Maintainable Pixlr integration	Pixlr is integrated through a dedicated adapter component. This keeps the third-party dependency isolated and makes it easier to change or replace later.
Consistency for image upload	The upload process is CP-oriented: an image is only published after the file and metadata are successfully stored. Errors lead to FAILED status or compensating cleanup.

5. Building Block View

The building block view shows the static decomposition of PixelShare. It uses three levels: the overall system, the internal backend decomposition and the detailed core-domain image module.

5.1 Level 1 - Whitebox Overall System



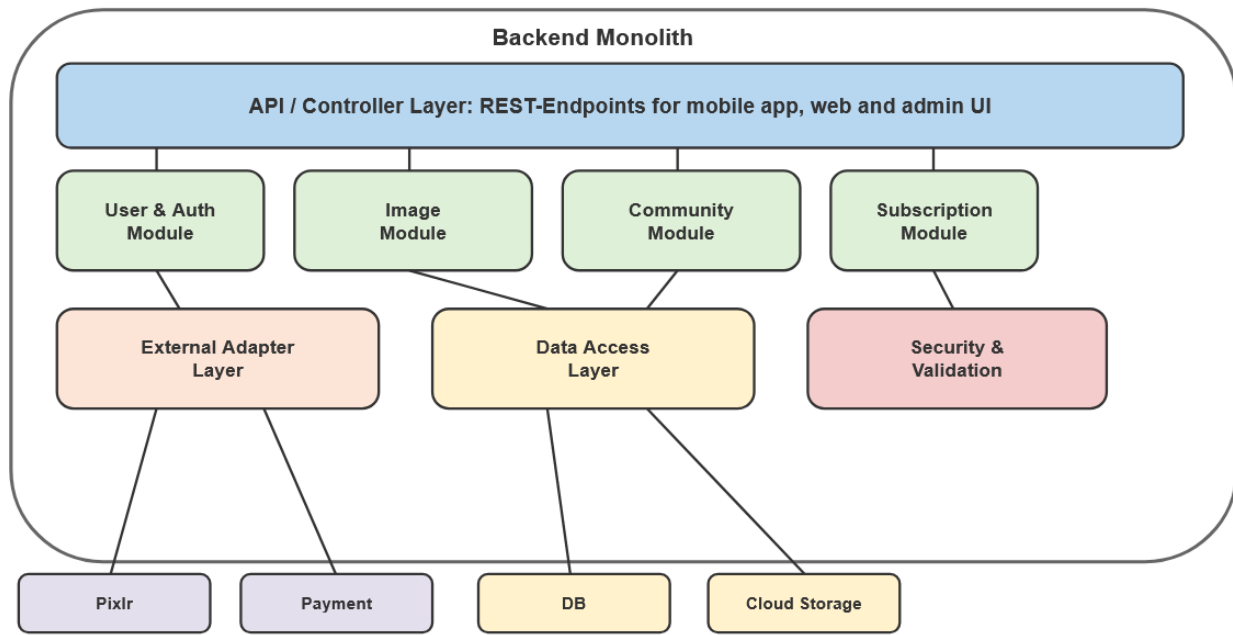
Building Block Level 1

Motivation

Level 1 shows the main building blocks of PixelShare and the most important external systems. This view is intentionally abstract, so that non-technical stakeholders can understand the overall system without implementation details.

Building Block	Responsibility
Mobile App	iOS/Android client for end users. Handles user interaction, upload selection, feed navigation, community actions and local UI flows.
Web / Admin UI	Browser-based interface for administration and content moderation.
Backend Monolith	Central server-side application. Contains API endpoints, business logic, integration logic and access to persistence.
PostgreSQL Database	Stores structured data: users, profiles, image metadata, comments, likes, subscriptions, moderation states and audit data.
Object Storage	Stores large binary image files, edited image versions and thumbnails.
Pixlr	External image-editing provider accessed through the Pixlr Adapter.
Payment / App Store	External billing and subscription provider.

5.2 Level 2 - Whitebox Backend Monolith



Building Block Level 2

Motivation

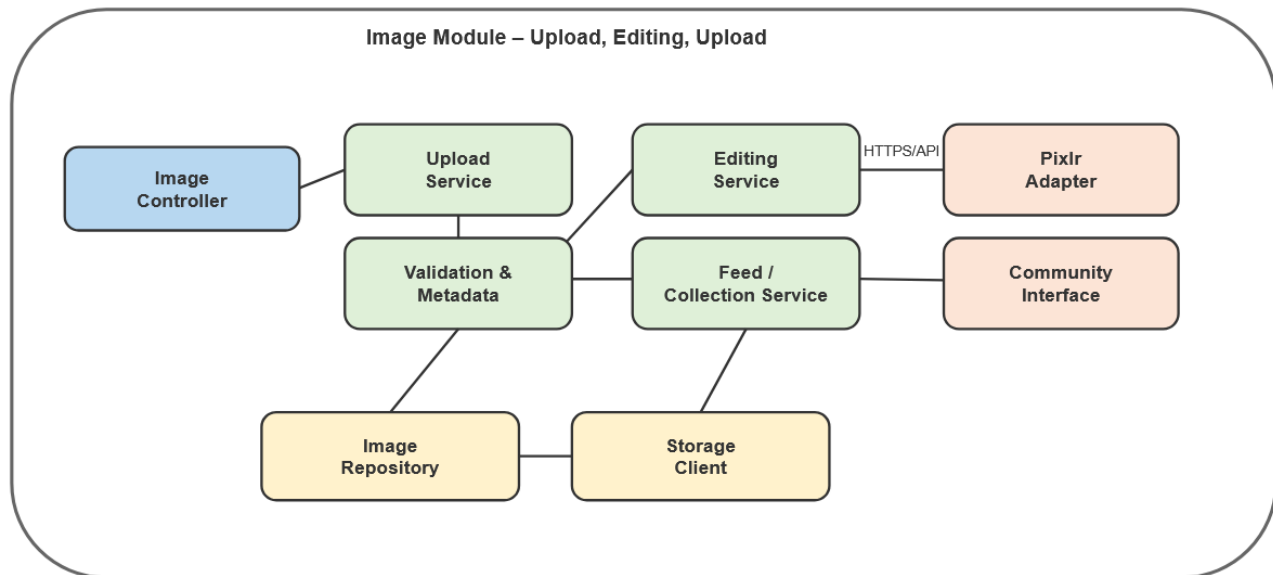
Level 2 decomposes the backend monolith into layers and modules. The backend remains one deployable unit, but the internal structure separates API handling, business logic, data access, security validation and external adapters.

Building Block	Responsibility
API / Controller Layer	Provides REST endpoints for the mobile app, web/admin UI and external callbacks. It validates request shape and delegates business work to modules.
User & Auth Module	Handles registration, login, user profiles, session/token handling and account-related business rules.
Image Module	Core-domain module for upload, validation, metadata, editing, publication, feed and collection integration.
Community Module	Handles likes, comments, follows, reports and community feed interactions.
Subscription Module	Handles subscription state, paid-plan access and communication with payment providers.
Security & Validation	Centralized authentication, authorization, role checks, input validation and protection of sensitive operations.
Data Access Layer	Encapsulates repositories and data access to PostgreSQL and metadata persistence.
External Adapter Layer	Encapsulates technical access to Pixlr, Payment/App Store and object storage. It isolates third-party API changes from business modules.

Important dependency rule

Dependencies flow from API to business logic and from business logic to data access or adapters. Business modules must not directly depend on provider-specific API details.

5.3 Level 3 - Whitebox Image Module



Building Block Level 3

Motivation

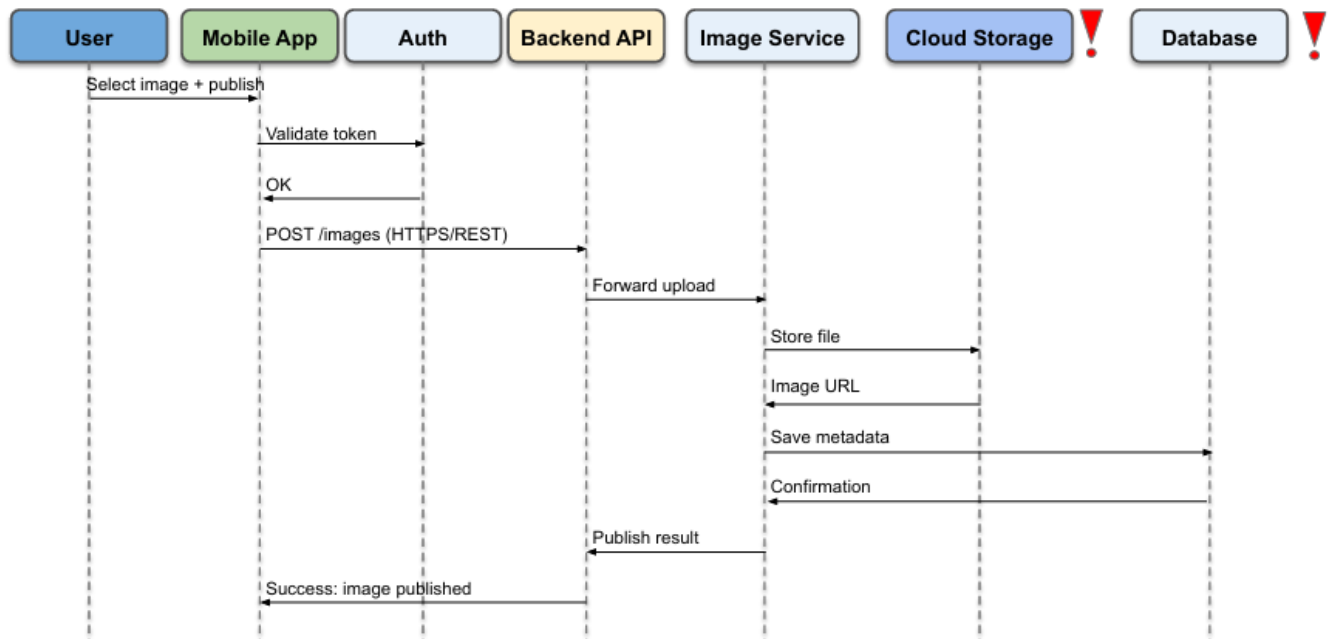
Level 3 zooms into the core domain of PixelShare: image upload, editing and publication. This area is architecturally important because it directly supports the main business value of the app.

Building Block	Responsibility
Image Controller	Receives upload, edit and publication requests from the API layer.
Upload Service	Coordinates file upload, metadata creation and publication state transitions.
Validation & Metadata	Checks file type, file size, ownership, visibility, title, description and tags.
Editing Service	Coordinates Pixlr editing sessions and stores edited image versions.
Feed / Collection Service	Makes published images available in feeds, profiles and collections.
Image Repository	Persists and reads image metadata from PostgreSQL.
Storage Client	Stores and retrieves image files from object storage.
Pixlr Adapter	Creates Pixlr edit sessions and handles edited-image callbacks.
Community Interface	Provides image-related hooks for likes, comments and reporting.

6. Runtime View

The runtime view documents two architecturally relevant scenarios: image upload/publication and image editing with Pixlr. They are relevant because they involve multiple internal components and external systems.

6.1 Runtime Scenario 1 - Upload and Publish Image



Runtime Scenario 1

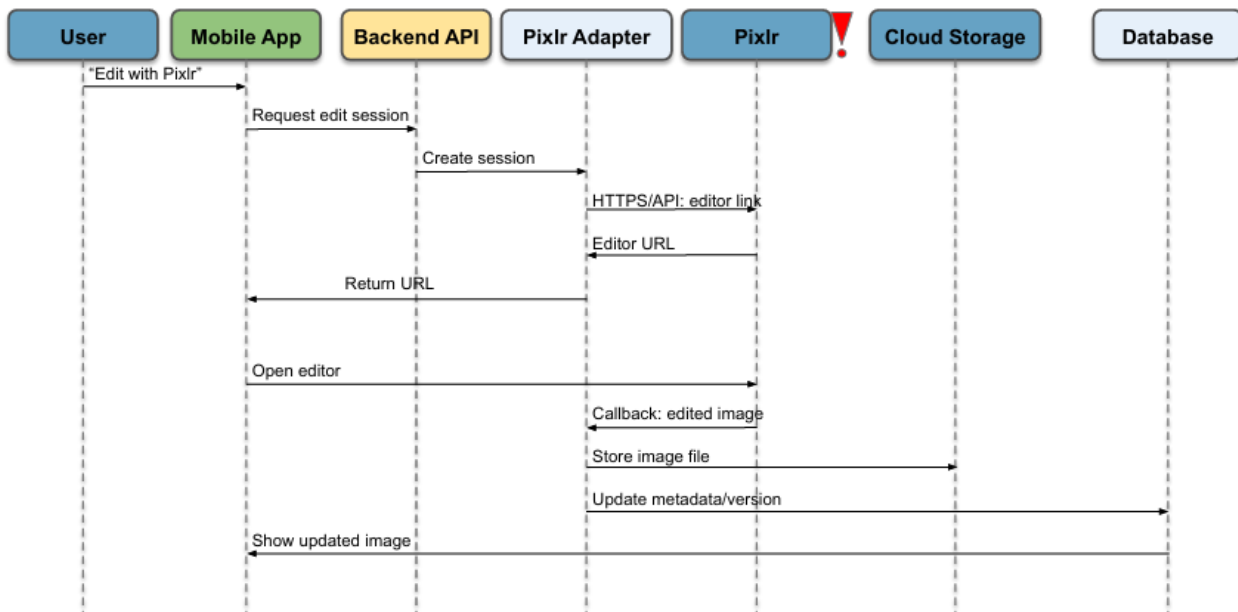
Description

1. The user selects an image in the mobile app and chooses to publish it.
2. The mobile app validates the token with the Auth component.
3. After successful authentication, the mobile app sends `POST /images` to the Backend API over HTTPS/REST.
4. The Backend API forwards the request to the Image Service.
5. The Image Service stores the file in Object Storage.
6. Object Storage returns an image URL.
7. The Image Service stores metadata in PostgreSQL.
8. After confirmation from the database, the publication result is returned to the mobile app.
9. The user sees that the image has been published.

Critical architectural aspects

- The scenario is CP-oriented in terms of the CAP discussion: consistency and partition tolerance are preferred over maximum availability.
- The image becomes visible only after file storage and metadata persistence both succeed.
- If storage or database persistence fails, the upload is not published. The system sets status `FAILED` or deletes the already stored file as compensation.

6.2 Runtime Scenario 2 - Edit and Save Image with Pixlr



Runtime Scenario 2

Description

1. The user chooses “Edit with Pixlr” in the mobile app.
2. The mobile app requests an edit session from the Backend API.
3. The Backend API delegates the request to the Pixlr Adapter.
4. The Pixlr Adapter creates a session with Pixlr through HTTPS/API.
5. Pixlr returns an editor URL.
6. The URL is returned to the mobile app and the editor is opened.
7. After editing, Pixlr sends a callback with the edited image.
8. PixelShare stores the edited image file in Object Storage.
9. PixelShare updates the metadata or image version in the database.
10. The mobile app displays the updated image.

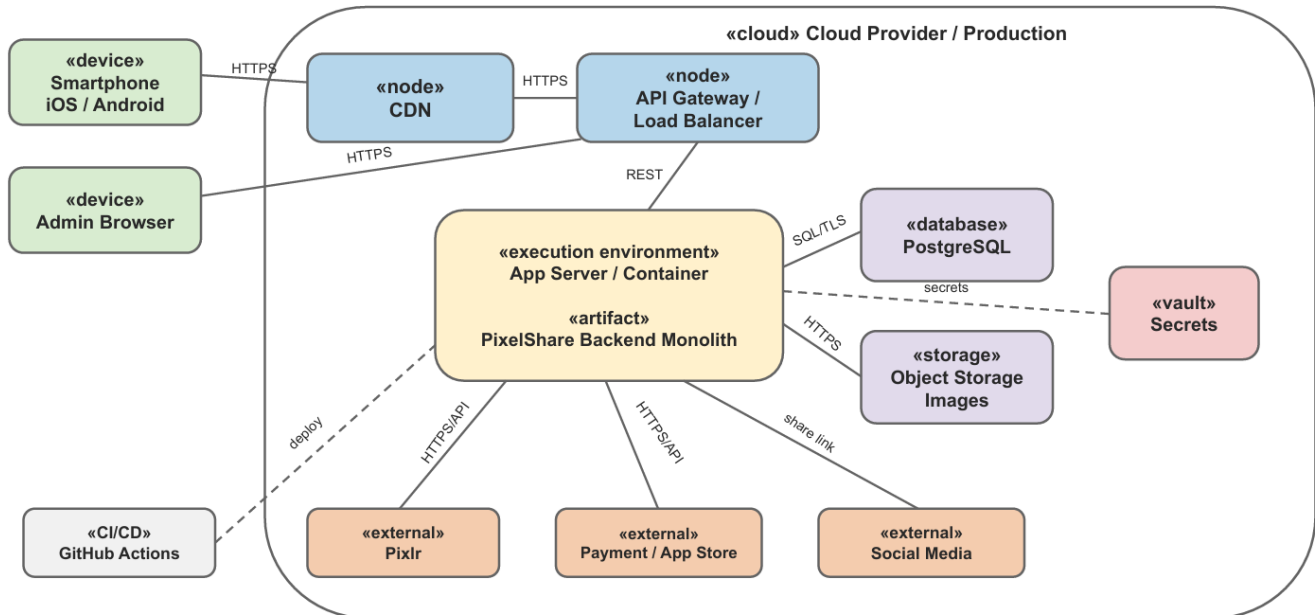
Critical architectural aspects

- Pixlr is isolated behind an adapter component.
- If Pixlr is unavailable, PixelShare remains usable for feed, profile, upload and community functions.
- Editing errors are displayed in an understandable way to the user.

7. Deployment View

7.1 Infrastructure Level 1

UML Deployment Diagram – PixelShare



Deployment Diagram

Motivation

The deployment view maps software building blocks to infrastructure elements. PixelShare uses a cloud-based production environment to support global mobile access, scalable image delivery, secure secrets management and repeatable deployments.

Quality and Performance Features

- The CDN improves global delivery of static content and images.
- The API Gateway / Load Balancer centralizes routing, authentication support, rate limiting and logging.
- The backend monolith runs in an app server/container and can later be scaled horizontally if needed.
- PostgreSQL stores structured data and must be backed up regularly.
- Object Storage stores large media files and reduces database load.
- Vault or a cloud secrets manager protects API keys, database passwords and provider credentials.
- CI/CD through GitHub Actions provides reproducible builds, tests and deployments.

Mapping of Building Blocks to Infrastructure

Software Building Block	Infrastructure Element
Mobile App	Smartphone device, distributed through iOS/Android app stores
Web / Admin UI	Admin browser, delivered over HTTPS
API / Controller Layer	App Server / Container behind API Gateway / Load Balancer

Software Building Block	Infrastructure Element
Backend Monolith	PixelShare backend artifact running in the app container
Data Access Layer	PostgreSQL database over SQL/TLS
Storage Client	Object Storage over HTTPS / S3-compatible API
Pixlr Adapter	External Pixlr system over HTTPS/API
Payment Adapter	Payment / App Store provider over HTTPS/API
Secrets Access	Vault / secrets manager
Deployment Pipeline	GitHub Actions

8. Cross-cutting Concepts

Cross-cutting concepts are system-wide rules that affect multiple modules and layers. They improve conceptual integrity and support the quality goals of maintainability, security, reliability and performance.

8.1 Consistent Development and Documentation Process

Goal

Ensure that all developers work with the same process and documentation rules.

Rules

- Use a shared Git workflow with pull requests.
- Perform code reviews before merging into the main branch.
- Document relevant architecture decisions as ADRs.
- Keep diagrams and arc42 documentation aligned with implementation changes.
- Apply KISS and YAGNI to avoid over-engineering for the MVP.

Affected areas

All developers, all modules, all architecture decisions and the CI/CD pipeline.

8.2 Layered Architecture with Clear Responsibilities

Goal

Keep the monolith maintainable by enforcing a clear internal structure.

Rules

- Presentation/API layer handles incoming requests and responses.
- Business Logic layer contains business rules for upload, comments, likes, subscriptions and moderation.
- Data Source layer encapsulates database, object storage and external technical integrations.
- Dependencies flow downwards only.
- New features follow the same structure.

Affected areas

Backend monolith, API endpoints, business modules, repositories and adapter components.

8.3 Security, Authorization and Privacy

Goal

Protect user accounts, images, subscription data and admin functions.

Rules

- Users must log in before uploading images, commenting or using protected functions.
- Admin and moderation operations require role-based authorization.
- Passwords are stored encrypted/hashed, never in plain text.
- All communication with external systems uses HTTPS/TLS.
- Sensitive keys and credentials are stored in Vault or a cloud secrets manager.
- Personal data and images must comply with GDPR/privacy requirements.

Affected areas

Login, upload, comments, subscription, admin functions, Pixlr integration, payment integration, cloud storage and APIs.

8.4 Error Handling and Resilience

Goal

Keep user-visible behavior understandable and avoid inconsistent system states.

Rules

- Technical errors are logged with context, but user messages remain understandable.
- Pixlr outages must not make the whole app unavailable.
- Upload failures must not create published images without valid metadata.
- Cloud storage and database failures lead to a failed upload state or compensating cleanup.
- External adapters handle provider-specific error responses.

Affected areas

Image upload, editing, feeds, Pixlr adapter, payment adapter, support diagnostics and monitoring.

8.5 Configuration and Secrets Management

Goal

Separate configuration from code and protect sensitive operational data.

Rules

- Environment-specific configuration is provided through deployment configuration.
- Secrets are injected securely at runtime through Vault or a secrets manager.
- No passwords, API keys or certificates are committed to source code.
- Configuration changes must be documented and reviewed.

Affected areas

Backend deployment, database connection, Pixlr API credentials, payment credentials, CI/CD pipeline and production operations.

9. Architecture Decisions

ADR-001: Use a Monolithic Backend with Layered Internal Structure

- **Status:** Accepted
- **Context:** The MVP must be implemented with a small team, limited budget and short timeline.
- **Considered alternatives:** Monolith, SOA, microservices.
- **Decision:** PixelShare uses a monolithic backend with a layered internal structure.
- **Rationale:** A single deployable unit is easier to develop, test and operate for a start-up MVP. Microservice complexity is not necessary at the beginning.
- **Consequences:** Development remains faster and simpler. The main risk is tight coupling; therefore clear packages, layers, code reviews and refactorings are required.
- **Related items:** [C_1](#), [C_2](#), [C_4](#), [QG-1](#), [QG-3](#), [Risk R-1](#).

ADR-002: Store Image Files in Object Storage and Metadata in PostgreSQL

- **Status:** Accepted
- **Context:** PixelShare stores many image files and structured metadata for feeds, profiles, comments and likes.
- **Considered alternatives:** Store images directly in the database; store images on a local server; use object storage for files and a database for metadata.
- **Decision:** Image files are stored in object storage. PostgreSQL stores metadata, references, titles, tags, comments, likes and subscription data.
- **Rationale:** Object storage is suitable for large binary image files. PostgreSQL is suitable for structured relationships and queries.
- **Consequences:** The upload flow must handle consistency between object storage and database. Failed uploads require a failed state or compensating cleanup.
- **Related items:** [QG-1](#), [QG-2](#), [Runtime Scenario 1](#), [ADR-004](#), [Risk R-4](#).

ADR-003: Integrate Pixlr Through a Dedicated Adapter Component

- **Status:** Accepted
- **Context:** PixelShare integrates Pixlr for external editing functions. Third-party APIs may change or be temporarily unavailable.
- **Considered alternatives:** Direct API calls in business logic; dedicated adapter component; build an own editing tool.
- **Decision:** Pixlr is accessed through a dedicated Pixlr Adapter.
- **Rationale:** The adapter isolates provider-specific details and keeps the business logic cleaner. It also makes future replacement easier.
- **Consequences:** Additional adapter code and tests are required. Pixlr-specific errors are handled centrally.
- **Related items:** [Cross-cutting Concept 8.4](#), [Runtime Scenario 2](#), [Risk R-3](#).

ADR-004: Prefer CP-oriented Upload Consistency for Image Publication

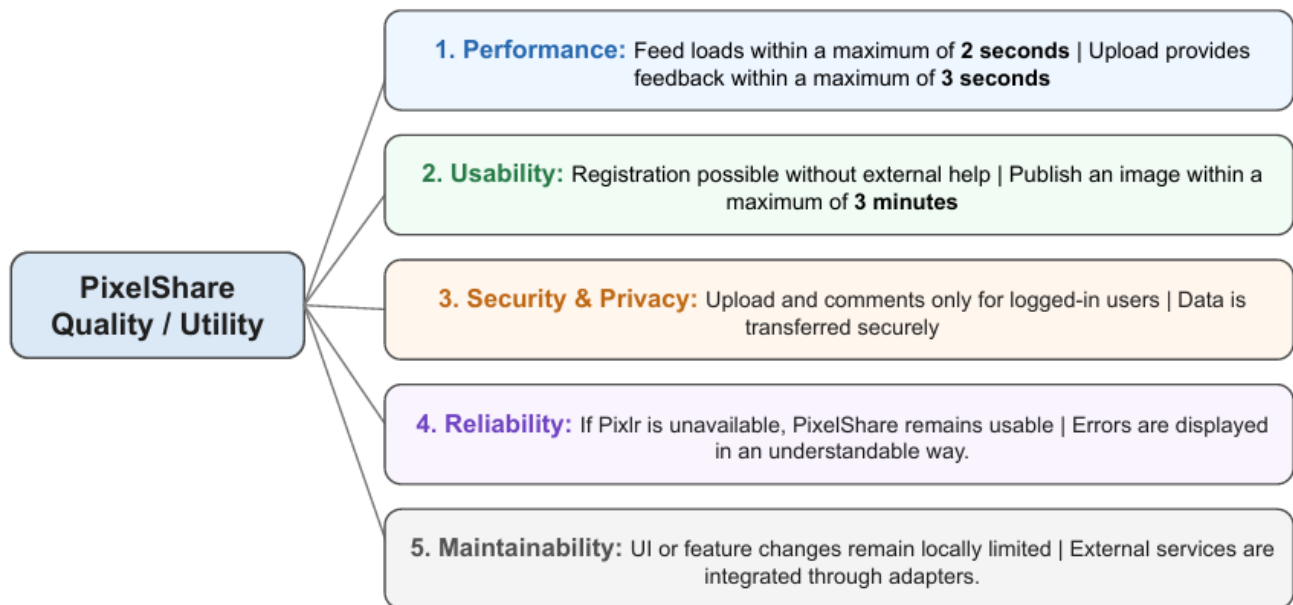
- **Status:** Accepted
- **Context:** Image upload involves object storage and database metadata. A network or cloud failure can otherwise create inconsistent data.
- **Considered alternatives:** Always accept upload and repair later; block publication until file and metadata are consistent; use a complex distributed transaction.
- **Decision:** PixelShare uses a CP-oriented upload flow: consistency and partition tolerance over maximum availability.
- **Rationale:** A broken or incomplete image in the feed is worse than a clear upload error. Users can retry the upload.
- **Consequences:** Upload may be temporarily unavailable during storage/database problems. The system uses PENDING, PUBLISHED and FAILED states and compensating cleanup.
- **Related items:** [Runtime Scenario 1](#), [Deployment View](#), [Quality Scenarios QS-2](#) and [QS-7](#).

ADR-005: Use API Gateway / Load Balancer, CDN and CI/CD Pipeline

- **Status:** Accepted
- **Context:** PixelShare needs secure and reliable access for mobile users and repeatable deployments.
- **Considered alternatives:** Direct backend exposure; backend behind API gateway/load balancer; manual deployment; CI/CD deployment.
- **Decision:** PixelShare is deployed behind an API Gateway / Load Balancer, uses a CDN for static/media delivery and GitHub Actions for CI/CD.
- **Rationale:** This improves routing, security, reliability and operational repeatability.
- **Consequences:** Deployment configuration becomes an important operational asset. Secrets management and monitoring must be handled carefully.
- **Related items:** [Deployment View](#), [Cross-cutting Concepts 8.3](#) and [8.5](#).

10. Quality Requirements

10.1 Quality Tree



Quality Tree

10.2 Quality Scenarios

ID	Quality Attribute	Scenario	Response Measure
QS-1	Performance	A user opens the Home Feed under normal conditions.	The first visible images and posts load within 2 seconds.
QS-2	Performance	A user uploads an image under normal network conditions.	The app shows upload progress or feedback within 3 seconds.
QS-3	Usability	A new user starts with no prior knowledge of PixelShare.	The user registers, uploads and publishes an image with title and description within 3 minutes without external help.
QS-4	Security & Privacy	A non-logged-in user tries to upload an image or write a comment.	The system prevents the action immediately and asks the user to log in.
QS-5	Reliability	Pixlr is unavailable during an edit attempt.	PixelShare remains usable for feed, profile, upload and comments; the user sees a clear editing error.
QS-6	Maintainability	The Pixlr API changes or must be replaced.	Only the Pixlr Adapter and its tests require changes; core image business logic remains stable.
QS-7	Reliability / Consistency	Object storage succeeds but database metadata persistence fails.	The upload is not published; the system sets status FAILED or deletes the stored file as compensation.

ID	Quality Attribute	Scenario	Response Measure
QS-8	Security	Admin functionality is requested by a normal user.	The system rejects the request and logs an authorization event.

10.3 Traceability Matrix

Quality Goal	Supporting Decisions / Concepts	Related Runtime or Deployment View
Performance and Reliability	ADR-002 Object Storage, ADR-005 CDN/API Gateway, Cross-cutting Concept 8.4 Error Handling	Runtime Scenario 1, Deployment View
Privacy and Security	ADR-005 API Gateway, Cross-cutting Concept 8.3 Security, C_5 Privacy and Secure Communication	Runtime Scenario 1, Deployment View
Usability and Learnability	Simple mobile flow, readable error messages, Cross-cutting Concept 8.4	Runtime Scenario 1 and Runtime Scenario 2
Maintainability	ADR-001 Layered Monolith, ADR-003 Pixlr Adapter, Cross-cutting Concept 8.2	Building Block Level 2 and Level 3
Consistency	ADR-004 CP-oriented Upload Consistency	Runtime Scenario 1, Deployment View

11. Risks and Technical Debt

R-1 Monolith Becomes Too Tightly Coupled

- **Description:** If modules are not clearly separated, future changes become difficult and error-prone.
- **Measure:** Use clear packages and layers, define architecture rules, perform code reviews and plan regular refactorings.

R-2 Performance Problems During Growth

- **Description:** Image upload or the feed may become heavily loaded while the entire monolith has to be scaled.
- **Measure:** Use object storage, caching, monitoring and later extract individual functions if necessary.

R-3 Dependency on Pixlr

- **Description:** Outages or API changes at Pixlr can affect the image-editing functionality.
- **Measure:** Use an adapter component, fallback behavior, clean error handling and a documented interface.

R-4 Storage / Database Inconsistency

- **Description:** Image file storage and metadata persistence can fail independently.
- **Measure:** Use PENDING/PUBLISHED/FAILED states and compensating cleanup for failed uploads.

R-5 Privacy and Compliance Gaps

- **Description:** Personal data, images and subscription data are sensitive.
- **Measure:** Apply secure communication, role checks, encrypted password storage, secrets management and GDPR/privacy review.

R-6 App Store Approval Delay

- **Description:** iOS/Android review processes can delay release.
- **Measure:** Prepare app store requirements early, document privacy behavior and keep release candidates testable.

12. Glossary

Term	Definition
PixelShare	Mobile image-sharing application for photography enthusiasts.
MVP	Minimum Viable Product; the first usable product version with the most important features.
Monolith	Architecture style where the backend is deployed as one application unit.
Layered Architecture	Internal structure separating API/presentation, business logic and data source concerns.
Backend API	Server-side REST interface used by mobile and admin clients.
Object Storage	Cloud/file storage for large binary objects such as images and thumbnails.
Metadata	Structured information about images, for example title, tags, owner, URL and visibility.
PostgreSQL	Relational database used for structured PixelShare data.
Pixlr	External image-editing provider integrated into PixelShare.
Adapter	Component that encapsulates access to an external system and hides provider-specific details.
CDN	Content Delivery Network used to deliver cached content closer to users.
API Gateway	Infrastructure component that routes and protects incoming client requests.
Authentication	Verification of the identity of a user.
Authorization	Verification that an authenticated user is allowed to perform an action.
GDPR	General Data Protection Regulation; European privacy regulation relevant for personal data processing.
CI/CD	Continuous Integration / Continuous Delivery or Deployment; automated build, test and deployment pipeline.
Runtime Scenario	Description of how system components communicate during a concrete use case.
UML Deployment Diagram	Diagram that maps software artifacts to technical infrastructure nodes.
Quality Scenario	Concrete and measurable scenario that describes how a quality attribute is evaluated.
CP-oriented	CAP-related decision prioritizing consistency and partition tolerance over maximum availability.